

Spatio- temporal modelling of soil moisture in Taylor Valley, Antarctica.

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Abstract

Soil moisture is one of the key drivers of biodiversity and community structures in Taylor Valley, part of the McMurdo Dry Valleys, Antarctica's largest ice-free environment. Its spatial distribution is controlled by complex hydrological and atmospheric drivers. The resulting spatial heterogenous dynamics cannot be obtained through field-measurements only. Spatio-
10 temporal modelling approaches are therefore required to assess soil moisture dynamics in Taylor Valley at landscape scale. As machine learning algorithms were successfully used for comparable problems, we used a Random Forest algorithm based approach, integrating continuous in situ soil moisture measurements from a network of 18 microclimate sensor stations operating during the 2021–2023 austral summers, providing high temporal resolution training data, and high-resolution multispectral PlanetScope imagery, alongside the topographic wetness index and modelled potential global radiation. Doing
15 so, we were able to generate a model being able to predict soil moisture in Taylor Valley spatio-temporally, with promising model performance.

1 Introduction

Soil moisture is one of the key drivers of biodiversity and community structures in Antarctic terrestrial ice-free environments (Fell et al., 2006; McKnight et al., 1999; Treonis et al., 1999). Its spatial and temporal variability is strongly influenced by
20 biogeochemical and physical ecosystem characteristics, which show pronounced heterogeneity across the McMurdo Dry Valleys (MDVs) (Barrett et al., 2004; Bockheim and McLeod, 2008). As Salvatore et al. (2023) have reported, the drivers of soil moisture in the MDVs, and Taylor Valley in particular, are multiple and complex.

The MDVs represent the largest ice-free area in Antarctica, with a combined area of approximately 4800 km² (Doran et al., 2002). The annual precipitation does not exceed 50 mm water equivalent, thus classifying the MDVs as a polar desert (Fountain
25 et al., 2010). The Taylor Valley is one of the MDVs and has a high density of soil water content influencing properties (Salvatore et al., 2023).

The mean annual air temperature in Taylor Valley ranges from approximately -23°C to -15°C, while the mean annual soil surface temperature is approximately -19°C. However, during the austral summer months (November-February), the soil

temperature can exceed 0°C, reaching temperatures in excess of 20°C in extreme cases (Doran et al., 2002). A particularly strong precipitation gradient has been identified in Taylor Valley (Fountain et al., 2010).

Soil types in Taylor Valley vary considerably, but are predominantly characterised by sandy, silty substrates with gravel, and an extremely low organic matter content. Variations in the degree of cryoturbation exhibited by these soils are also observed. Soils affected by cryoturbation (Typic Haploturbels) occur in the eastern sector of the Valley, near the Ross Sea, and non-cryoturbated soils (Typic Anhyorthels) occur on the west side of the valley, in closer proximity to the Transantarctic Mountains, within regions characterised by dry frozen permafrost. The distribution of salt occurrences across the valley exhibits significant variation, manifesting predominantly as patches, and constituting less than 10% of all observed soil types (Bockheim and McLeod, 2008; Levy et al., 2012).

Main drivers of soil moisture are wetting through nearby overland flow due to seasonal glacial streams (McKnight et al., 1999; Gooseff et al., 2003b), melting of snow patches or glaciers (Eveland et al., 2013), deposition of atmospheric water vapour through hygroscopic salts in the soil (Levy et al., 2012), as well as snowfall (Myers et al., 2022) or capillary moisture transfer from melting buried ground ice (Fountain et al., 2014).

Conventionally, the measurement of soil moisture in Taylor Valley was conducted through field-collected samples (Barrett et al., 2004; Barrett et al., 2008; Treonis et al., 1999) or in-situ measurements in the field (Levy et al., 2011). There have been only a limited number of attempts to broaden the soil moisture assessment using remote sensing methods for spatio-temporal modelling in Taylor Valley to overcome the limitations of pointwise non-continuous measurements. However there is an increasing interest in this information (Salvatore et al., 2023; Thapa-Magar et al., 2025).

Recently a reflectance-based linear modelling approach retrieving the soil moisture was presented by Salvatore et al. (2023). The approach utilised multispectral reflectance relationships of the soils to their moisture measured in the laboratory. Despite increasing interest in soil moisture dynamics in Taylor Valley no model integrating field-based measurements and spatio-temporal predictors has been developed so far. This gap becomes particularly evident when considering the lack of multi-predictor soil moisture models for this region. As previously mentioned, the moisture in Taylor Valley is driven by complex relationships.

Given the non-linear and interacting controls of soil moisture, machine learning (ML) approaches provide a suitable modelling framework. In this context, the relationship between the target variable (i.e. soil moisture) and the predictors will be learned based on a training dataset (Lary et al., 2016). The Random Forest algorithm (Breiman, 2001) is widely used in the field of ecology and has been successfully used to model soil moisture in a variety of environments (Shahriari et al., 2025; Singh et al., 2023).

The objective of this study is to use this algorithm to model the soil moisture in the challenging and unique environmental conditions of the MDVs with field-based training data. Contrary to the approach taken in preceding studies, which relied on laboratory-derived soil moisture data, the present study utilised a sensor network comprising 18 sites with Temperature-Moisture-Sensors (TMS4) installed in Taylor Valley from the 2021/2022 to the 2022/2023 austral summer season. These sensors recorded measurements at 15-minute intervals, thereby providing a unique, high-resolution dataset of continuous in

situ soil moisture. The TMS4 sensors were designed with challenging environments in mind (Wild et al., 2019). This dataset enables the development of field-based models and the exploration of spatial and temporal soil moisture patterns across Taylor Valley.

To predict soil moisture across the landscape, the modelling approach incorporates spatially available variables that reflect the key controls of soil moisture distribution. Multispectral data has been previously used successfully to assess optical soil properties driven by differences in soil moisture (Salvatore et al., 2023), and PlanetScope provides this data with a high spatial and temporal resolution (Planet Labs, 2025). In addition, topographic variables derived from a digital elevation model can account for slope and drainage (Sørensen et al., 2006), and estimates of incoming solar radiation to reflect energy inputs controlling soil thaw and evaporation (Western et al., 1999). Based on these data sources, the following research questions were addressed.

1. Is it possible to measure soil moisture with this type of sensor in this environment? Can significant differences in soil moisture be detected on a small scale?
2. Is it possible to fit a useful and meaningful random forest model based on satellite derived imagery, potential radiation and topographic predictors that individually show a significant relationship with the soil moisture measurements in eastern Taylor Valley?

2 Methods

A schematic overview of the workflow applied to examine the study’s objectives is presented in Figure 1. Taylor Valley was selected as the exemplary study area due to its unique environmental conditions, and the availability of a data set comprising multiple soil moisture measurements across the valley.

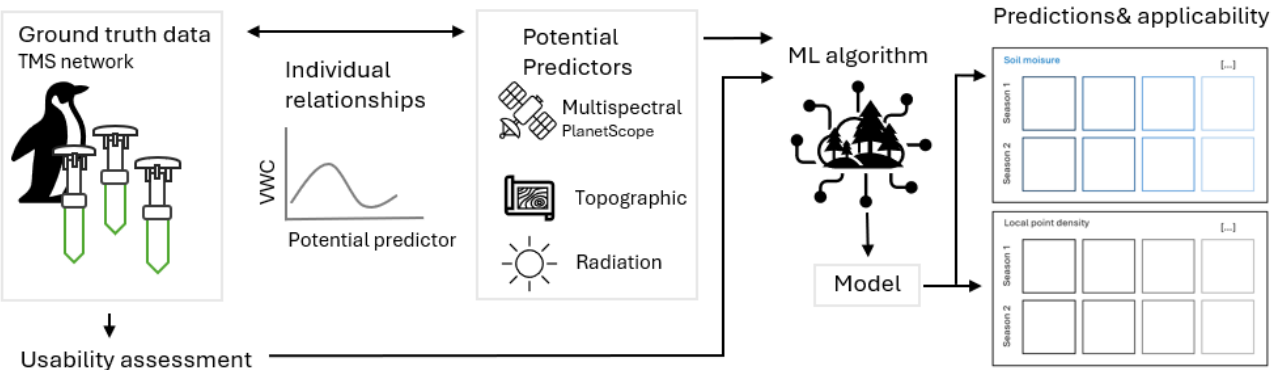


Figure 1 Schematic overview of the planned analyses and model framework.

2.1 Software

85 All statistical analyses and modelling attempts were conducted using the R software environment, version 4.4.2 (R Core Team, 2024), unless claimed differently.

2.2 Soil moisture ground truth data

The used soil moisture data was from a network placed in Antarctica for the 2021/2022 until the 2022/2023 austral summer season. An overview of the placement of the loggers is shown in Figure 2. According to the sensor network concept description (Valdes,2021; supplementary material), the loggers were placed on the 24th of December 2021 in front of Canada Glacier (Plot 2) and Commonwealth Glacier (Plot 4), providing a gradient across eastern Taylor Valley. Within the plot areas (1 km²), sub-plots of three moisture levels were identified via SWIR reflectance bands (30 m²). At Site (sub-sub-plot) scale, the elevation on one meter level was chosen to find a gradient for the placement of the 1 m² sites. Two to three sensors were placed on site scale, resulting in 50 sensors in total placed on 18 sites. An overview of sensors per site, including the soil moisture measurements, can be found in the supplementary material (S1). Therefore, this data set can represent a gradient of soil moisture influencing variables across Taylor Valley. TMS loggers measure soil moisture with the widely used time-domain-transmission (TDT) method, with a sensitivity of 0.01. With the TDT method, soil moisture measurements are only possible in non-frozen soil (Wild et al., 2019). Periods of frozen soil can be detected with the soil temperature sensor (Man et al., 2023) and were excluded from the dataset. This led to remaining periods of available soil moisture measurements from end of December 2021 to mid-February 2022 for the first and mid-November 2022 to beginning of December 2022 (Plot 2 and 4), respectively beginning of January (Plot 4 only). The loggers of Plot 2 were already retrieved in December 2022, while the loggers of Plot 4 remained in the field until January 2023. Additionally, loggers which were exposed from the soil during the measuring period were excluded, as recommended by Wild et al. (2019). If an indication for exposure of the sensor was present due to in-field observations and the logger measurements on sub-site level showed a significant difference according to the model described in the usability assessment section, the logger was excluded. Further information on the identification of exposed sensors and the influence of sensor exposure on VWC measurements can be found in the supplementary material (S2). To retrieve the volumetric water content (VWC), calibration is needed. While there are calibration curves provided for most common soil types (Wild et al., 2019), custom calibration is recommended for special environments (Wild et al., 2019). As mentioned by Bockheim and McLeod (2008), the soil types vary across Taylor Valley. The bulk density and soil type is needed to generate a custom calibration. As these parameters are not available, it was tested whether the universal calibration, presented by Kopecký et al. (2021), would be suitable for this case (for more detailed information, see supplementary material S3). For these reasons, the universal calibration was used to calibrate the VWC. Data preparation of the soil moisture data was

conducted with the ‘myClim’ package (Man et al., 2023). For the remaining measurements, the daily mean was calculated to be used as training data for the machine learning modelling approach.

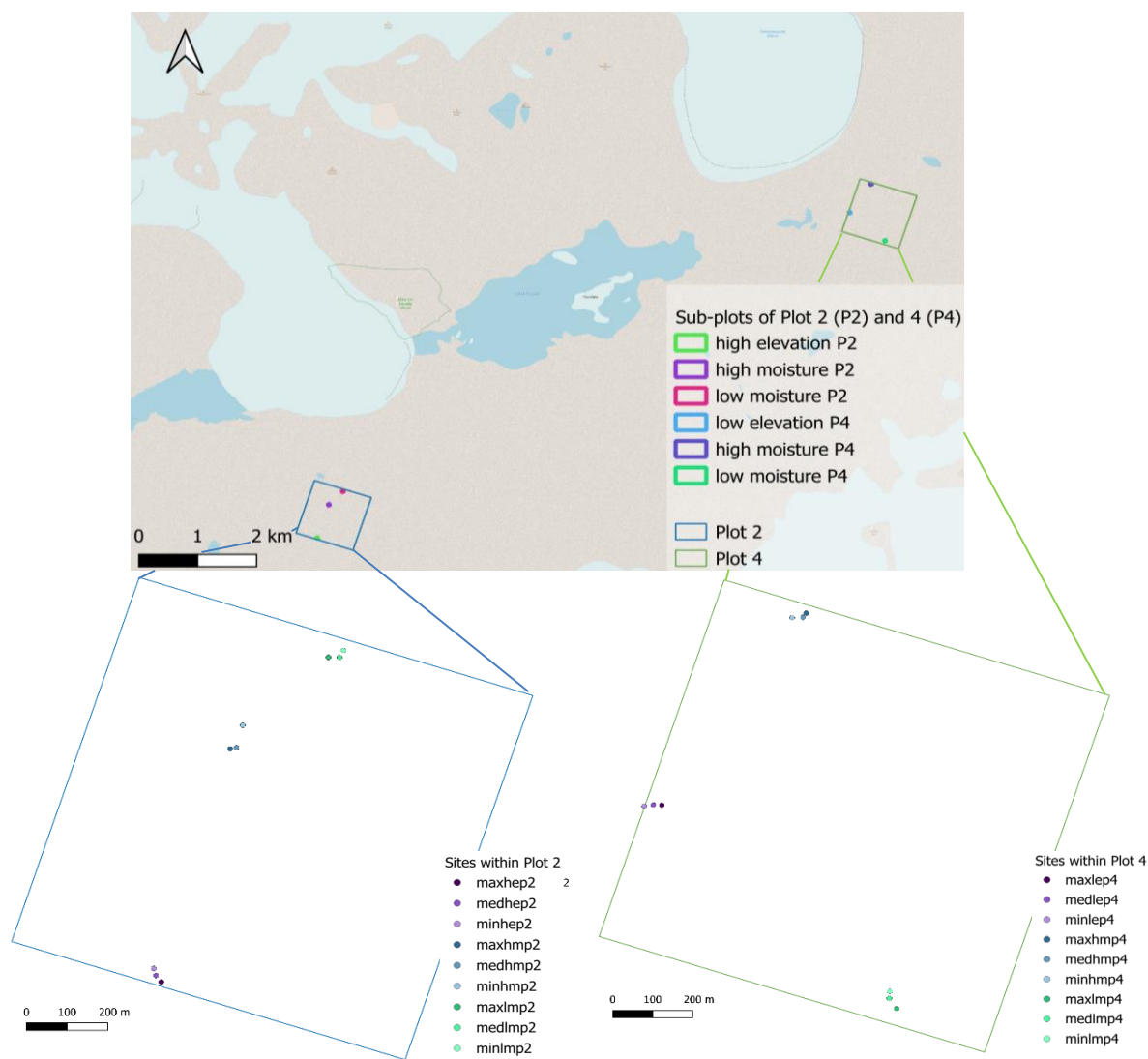


Figure 2 Overview of the nested design of logger placement in Taylor Valley. The presented sites are the smallest geo-referenceable site level. Two to three loggers are placed per these sites, but not shown in this overview.

115 **2.2.1 Usability assessment**

Prerequisites for the soil moisture data to be used as training data in the machine learning modelling approach are significant differences between the sites are visible and the variation of the soil moisture on sub-site level has to be significantly smaller than the variation between the sites to gain potentially meaningful model results. As the TMS sensors have not been used in

the extreme environment of the MDVs before, the prerequisites were assessed statistically prior to using them as training data.

120 VWC measurements were examined with generalized linear mixed models (GLMMs) with beta error distribution, because the VWC was distributed between zero and one, and a logit link function, using the implementation of the ‘glmmTMB’ package (Brooks et al., 2017). To assess logger- and site-specific effects, separate models were fitted for Plot 2 and Plot 4. Logger, site and their interaction were included as fixed effects and the week of the year as random intercept, to account for repeated measurements and temporal autocorrelation. The significance of fixed effects was evaluated using type III Wald chi-square

125 tests, implemented in the ‘car’ package (Fox and Weisberg, 2019). To quantify overall differences in soil moisture among all sites, a model including site as fixed effect and week of the year as random intercept was fitted. As no interaction was included in the model, the significance of the site effect on soil moisture was assessed with a type II Wald chi-square test in this case. Subsequently, estimated marginal means were calculated using the ‘emmeans’ package (Lenth, 2024) and back-transformed to the response scale for interpretability. Pairwise contrasts between logger identities within sites and pairwise contrasts

130 between sites were calculated using Tukey-adjusted comparisons. To evaluate whether variability resulting from logger differences within sites was smaller than variability between sites, the absolute values of the above described contrasts were tested for significant differences using a Kruskal–Wallis rank sum test (Kruskal and Wallis, 1952), followed by Dunn’s test for multiple pairwise comparisons (Dunn, 1964), as implemented in the ‘dunn.test’ package (Dinno, 2024, 2024).

2.3 Potential predictors

135 For every potential predictor, it was assessed whether a significant relationship between soil moisture and the predictor could potentially be assumed. This was done by training individual GLMMs structured similarly as described above and with the potential predictor as fixed effect and site as random intercept. The significance of the fixed effect was assessed with a type II Wald chi-square test. Only predictors with at least marginal significance ($p < 0.1$) were included and described as potential predictors below. Further information on the identification of potential predictors can be found in the supplementary material

140 (S4).

2.4 Topographic data

As it had the highest available spatial resolution (1 m²), the raster product of a LiDAR based digital elevation model (DEM), retrieved in 2014-2015, was used for topographic analyses (Fountain et al., 2017; OpenTopography, 2016). To account for topographic influences on soil moisture, slope, aspect, and the Topographic Wetness Index (TWI) were derived from the DEM.

145 Slope and aspect were calculated using the ‘terra’ package (Hijmans, 2024). The TWI was calculated according to the following formula:

$$TWI = \ln \frac{a}{\tan(b)} \quad (1)$$

Where ‘a’ represents the contributing upslope area, and ‘b’ is the slope. This was done using the SAGA wetness index algorithm (Boehner et al., 2001), implemented in the SAGA software environment version 9.7.2 (Conrad et al., 2015), and
150 accessed from R via the ‘qgisprocess’ package (Dunnington et al., 2024).

2.5 Global radiation

To account for seasonal variations in the global radiation (total irradiation [$W/(m^2 \cdot d)$]) (daily_glob_rad) and daily insolation time [h] (daily_insol), while taking topographic shading into account (Katurji et al., 2013), these predictors were calculated for every day in the measuring period where PlanetScope data was available, using the r.sun algorithm (Hofierka and Suri,
155 2007) from the GRASS GIS environment, version 7.8 (GRASS Development Team, 2020). The Linke atmospheric turbidity coefficient (Kasten, 1996) was set to 2.0, derived for the study area from Bason (2008) and Wendler and Ishikawa (1988).

2.6 Multispectral data

Multispectral data from the PlanetScope mission was selected because of the, with 3m per pixel and a nearly daily revisit, high spatial and temporal resolution (Planet Labs, 2025). Especially the high temporal resolution was a key criterion, as this
160 provided a higher chance of cloud-free scenes during the measuring period. It was crucial to use as many scenes as possible to provide as many as possible training data points. SuperDove and Dove-R sensors have been providing multispectral data since 2019, ongoing. Red, Green and Blue (RGB) and Near-Infra-Red (NIR) bands were selected due to their interoperability with Sentinel-2, which would be useful for further studies of longer time series in the future, as Sentinel-2 was launched in 2015 (ESA, 2025). Only images with less than 20% cloud-cover and complete coverage for the study area were included. The
165 PlanetScope Surface reflectance product (Collison and Curdoglo, 2025) was used. This led to sixteen available scenes, eleven for the 2020/21, and five scenes for the 2021/22 austral summer season (Planet Labs PBC, 2021, 2022, 2023). Additionally, six scenes were identified for December 2022 to January 2023, the period where data was only available for Plot 4.

2.7 Multispectral potential predictors

2.7.1 Distance to Snow/ Ice Patch

Snow is known to be an important driver of soil moisture in Taylor Valley (Eveland et al., 2013). Snow patches are normally derived from multispectral data with the SWIR based Normalized Difference Snow Index (NDSI) (Dozier, 1984; Rango and Itten, 1976). As PlanetScope data doesn’t provide a SWIR band, the NDSI couldn’t be used for the identification of snow/ice patches. Therefore, the Blue Snow Threshold approach, as introduced by Thaler et al. (2023), was used instead. As this algorithm had not been used in the Antarctic environment before, the suitability of the algorithm was assessed prior to the
175 predictor calculation, as described in the supplementary material (S5). Afterwards, the distance to the closest snow/ ice patch (snow_dist) was calculated for every pixel, using the ‘gridDist’ function from the ‘terra’ package (Hijmans, 2024).

2.7.2 Brightness Index

The Brightness Index (BI) can be used to account for soil moisture influences on soil colour (Escadafal, 1989; Mathieu et al., 1998). The BI was calculated as follows:

180
$$BI = \frac{\sqrt{red^2 + green^2}}{2} \quad (2)$$

To account for the change in soil moisture, the BI was not included directly, but as a pixel-wise difference to the pixel-wise mean BI over all scenes (BI_i_mean), and as the pixel-wise BI difference to the overall mean BI (BI_o_mean).

2.7.3 Normalized Difference Water Index

The Normalized Difference Water Index (NDWI), as introduced by McFeeters (1996), was calculated as follows:

185
$$NDWI = \frac{green - nir}{green + nir} \quad (3)$$

2.8 Machine learning model

As it is commonly used in environmental modelling, the machine learning algorithm Random Forest (Breiman, 2001) was chosen. In this study; the Random Forest implementation of the ‘randomForest’ package (Liaw and Wiener, 2002) was used and addressed via the ‘caret’ package (Kuhn, 2008). Model performance during the training steps was assessed with the root-mean-square error (RMSE). The final model performance metrics, RMSE, the mean absolute error (MAE) and the coefficient of determination (R^2), were assessed with the ‘global validation’ function implemented in the ‘CAST’ package (Meyer et al., 2024b).

2.8.1 Training and validation data

The daily mean VWC of all dates where PlanetScope scenes were available were matched with the corresponding potential predictors and used as training data. The period where only sensors from Plot 2 were available was used as an additional, independent test set, while being aware that, because of the nested network design, the conditions are not fully representative for the whole set up.

To address the issue of spatial autocorrelation, a 4-fold spatial cross-validation set up was used to split the training data. The Site Ids were used as splitting criteria, securing that training data from the same site cannot be found in the training and test set at once. Folds were created with the ‘createSpacetimeFolds’ function implemented in the ‘CAST’ package. The nearest neighbour distance distribution plot will be provided to control if the cross validation set up during model training can be considered as representative for the prediction situation. The distances in feature space, hence all predictors, will be considered. (Meyer et al., 2024a).

205 **2.8.2 Model training, tuning and prediction**

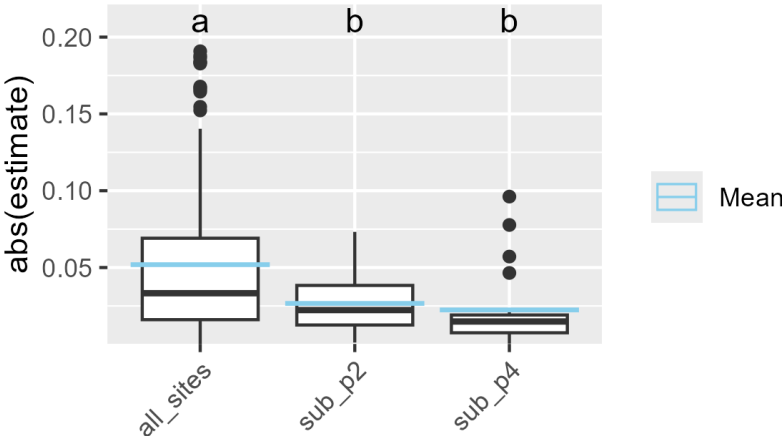
A spatial forward feature selection (FFS) (Meyer et al., 2019) was performed to further address the risk of overfitting. Due to computation time limitations, the number of randomly selected variables per split (mtry) was set to two during the FFS, and 50 trees were trained during this step. To fine-tune the model, the mtry value was tested from two to the number of, during the FFS selected, predictors. The optimal mtry value was chosen based on the smallest RMSE. During this step, the number of
210 trees was set to 500. Shapley values were calculated to assess the impact of individual predictors on the model output (Štrumbelj and Kononenko, 2014) using the ‘fastshap’ package (Greenwell, 2024).

The resulting model was used for spatial predictions of soil moisture for every available time step. Model predictions are only communicated within the area of interest (AOI), excluding the known lakes, ponds, and glaciers in Taylor Valley (Gardner, 2016), and within the area of applicability (AOA). As introduced and described by Meyer and Pebesma (2021), this is the area
215 where the model was able to learn about relationships based on the training data, and where the estimated spatial cross-validation performance is valid. In addition, the Local Point Density (LPD) will be presented. The LPD contains information on how many training data points with similar properties were included in the model training for each new prediction location (Schumacher et al., 2025).

3 Results

220 **3.1 Usability of TMS soil moisture measurements**

The variability of soil moisture was significantly different between sites compared to within sites ($p < 0.001$). Additionally, variability and mean VWC between all sites were significantly higher between sites than within sites (Fig.3).



3.2
Figure 3 Results of Dunn's test for comparison of absolute values of estimates between all sites and on sub-site level (sub), split by plots. Significant differences ($p < 0.001$) were indicated with different letters.
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The majority of pairwise site differences of soil moisture were significantly different ($p < 0.001$). While the mean VWC was ranging between 0.1 and 0.3, most of the sites had a mean VWC between 0.1 and 0.15 (Fig.4).

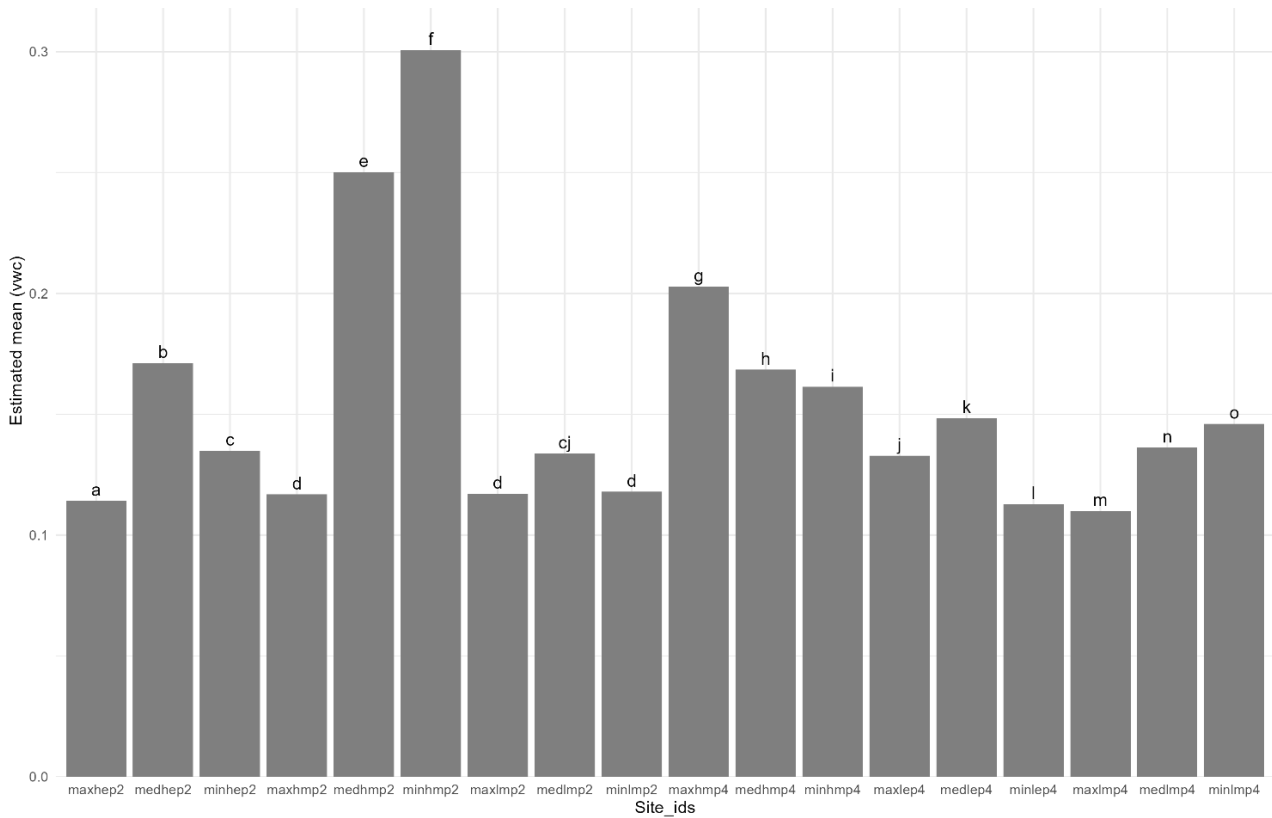


Figure 4 Pairwise contrasts between estimated mean VWC at all sites. Significant differences ($p < 0.001$) were indicated with the same letter.

3.2.1 Significant differences on sub-site level

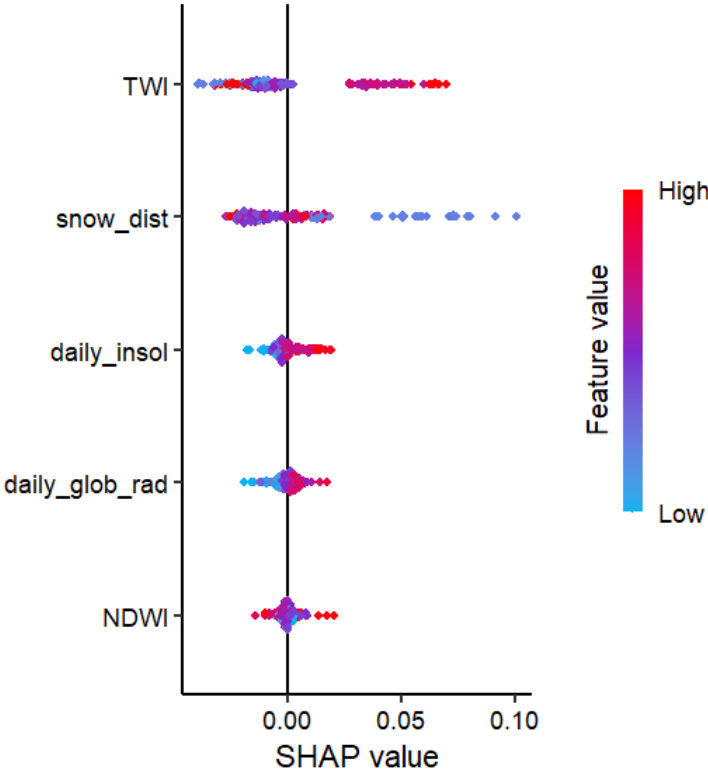
The soil moisture mean differed significantly ($p < 0.001$) on sub-site level between at least two loggers for the majority of sites. Detailed information about the differences can be found in the appendix (Fig.I, II).

3.3 Suitable predictors

Five predictors were selected during the FFS. The most important were the TWI and the distance to snow/ice patch (Tab.1). Shapley values, indicating the impact on the model output for individual predictors, are visualized in Figure 5. The results indicate that higher values of the TWI, potential global radiation and daily insolation time each lead to predictions of wetter conditions. Lower distances to snow/ice patches lead to predictions of wetter conditions as well. The Shapley values for the NDWI are not indicating an obvious impact pattern.

Table 1 Potential predictors, in order of importance, and their origin and temporal dynamics. During the FFS, for the final model, selected predictors are indicated and their importance for the model communicated.

Predictor	Temporal	Multispectral	Topographic	Selected for final model	Importance	Reference
TWI			X	X	46	(Boehner et al., 2001)
snow_dist	X	X		X	40	(Thaler et al., 2023)
daily_insol	X		X	X	14	(Hofierka and Suri, 2007)
daily_glob_rad	X		X	X	12	(Hofierka and Suri, 2007)
NDWI	X	X		X	6	(McFeeters, 1996)
BI_i_mean	X	X			-	(Escadafal, 1989; Mathieu et al., 1998)
BI_o_mean	X	X			-	(Escadafal, 1989; Mathieu et al., 1998)



245 **Figure 5** Shapley values (impact on model output) for the predictors included in the final model.

3.4 Model metrics

Based on the spatial cross validation, the final model had an RMSE of 0.042, a MAE of 0.029%, and the coefficient of determination (R^2) was 0.419. The performance evaluation with the additional test set led to an RMSE of 0.019, a MAE of

0.016% and a coefficient of determination of 0.419. While the R^2 was identical during both performance assessments, the RMSE and MAE indicated a better model performance for the additional test set. The nearest neighbour distance distributions (Fig.6) indicate that the cross-validation situation is not fully representative for the prediction tasks. Therefore, the model performance assessment may be slightly too optimistic.

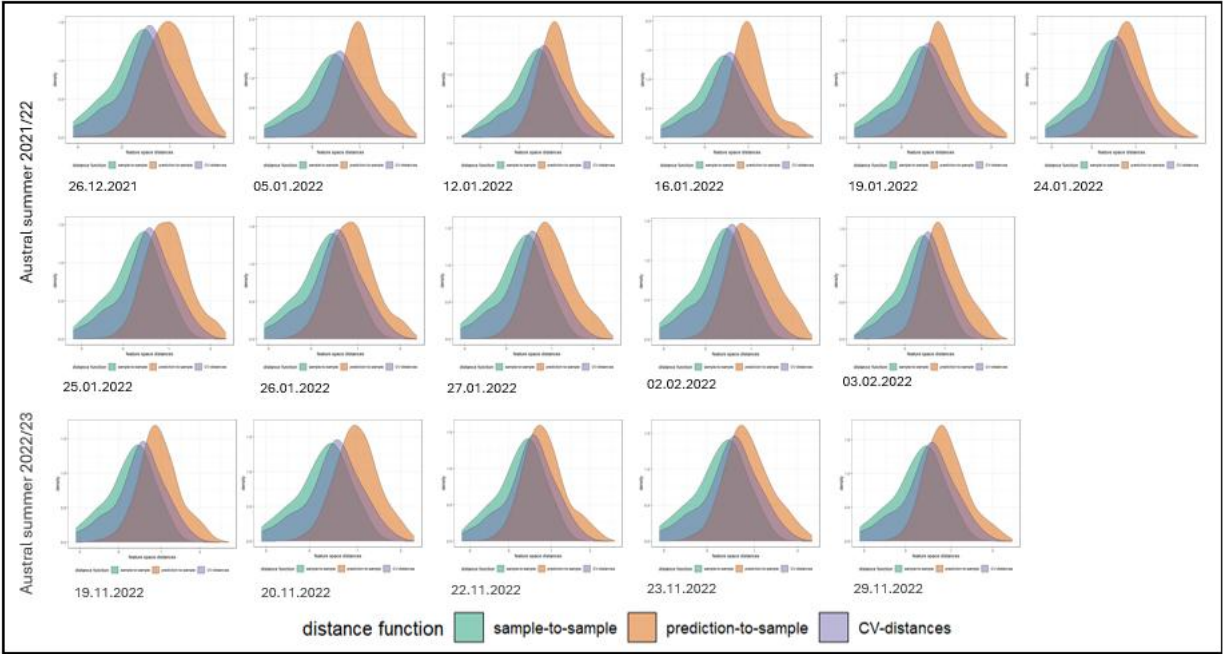


Figure 6 Nearest neighbour distance distributions. The y-axis represents the density, the x-axis the feature space distances. The cross-validation (CV)-distances in feature space are compared to the distances during the prediction task. Areas outside the area of interest and the area of applicability were excluded prior to the calculation.

3.5 Predictions

Model predictions for every time step included in the training data set are shown in Figure 7. Predictions are only communicated within the area of applicability and the area of interest. Different austral summer seasons are indicated as well. Due to the data availability, the time difference between time steps is varying. An animation of the spatial soil moisture predictions can be found in the supplementary material (S6) for a more detailed impression of the temporal soil moisture dynamics. The local point density for every prediction scene can be found in Figure 8.

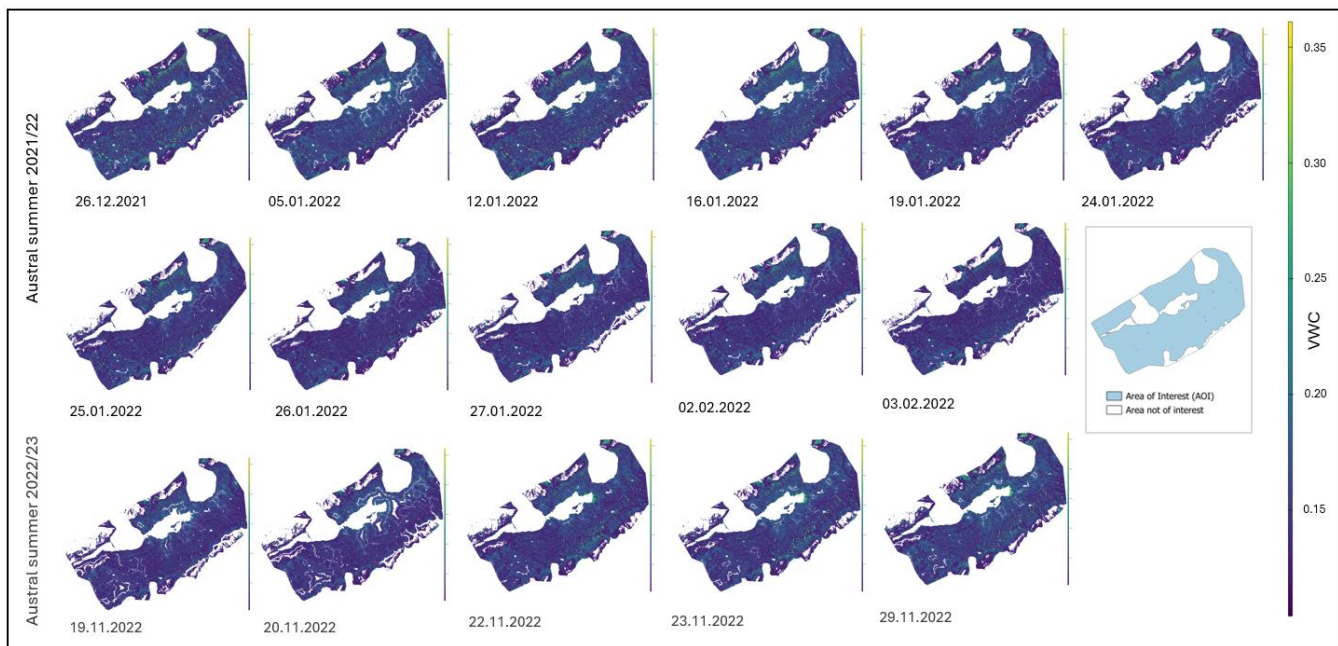


Figure 7 Predictions within the AOI and AOA for every time step included in the test set. The AOI is shown in light blue in the additional frame on the right. All additional white areas in the prediction plots are out of the AOA.

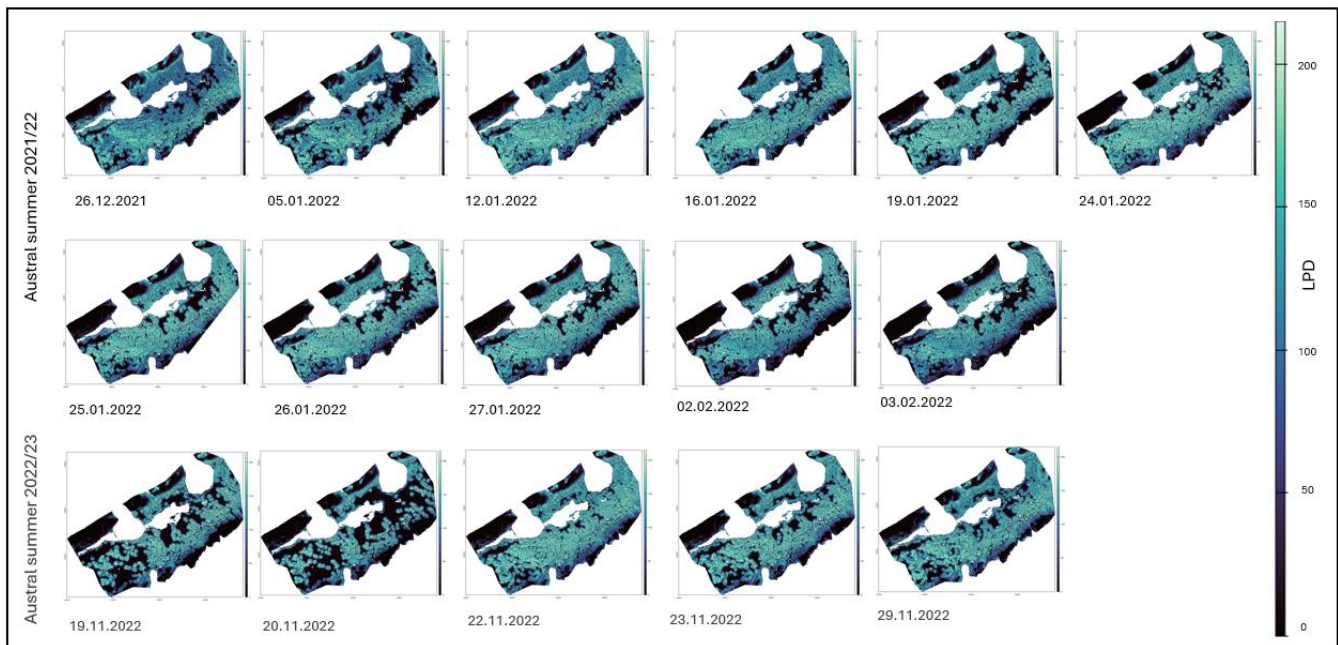


Figure 8 The LPD for every prediction scene. White areas are outside the area of interest.

4 Discussion

4.1 TMS Soil moisture data

270 The soil moisture differed significantly between sites (Fig.4), and the variance in soil moisture between sites exceeded the variance between individual loggers at the sub-site level (Fig.3). This is in line with Wild et al. (2019), who found that the TMS sensors can provide continuous measurements in other extreme environments, such as the Himalaya. Therefore, measuring soil moisture with the TMS loggers appears to be effective in the extreme environment of Taylor Valley, and the measurements provided valuable ground-truth data for the ML-modelling approach.

275 4.2 ML-modelling approach

The model performance metrics, MAE and RMSE, indicated a promising performance for spatio-temporal predictions of soil moisture in eastern Taylor Valley, while the coefficient of determination (0.419) suggests that the model does not fully capture the complexity of soil moisture drivers in Taylor Valley. The additional test set showed lower MAE and RMSE compared to the cross-validation. Since the additional test set contained measurements from plot 4 only, and the measurements represented a rather dry period, it cannot be considered as representative for the whole valley. Given that more training data from dry conditions was included in the model (Fig.4), this result was expected. Nonetheless, the performance assessment with the test set indicates that the model could be used for predictions beyond the measuring period, particularly under dry conditions.

280 The TWI was identified as the most important predictor in the model, followed by the distance to snow/ice patch as the second most important predictor (Tab.1). This is consistent with the findings of Gooseff et al. (2003a) and Ayres et al. (2010), who identified snow patches as an important source and influencing factor of soil moisture in Taylor Valley. The prominence of these two predictors as the most important is especially interesting in the context of Eveland et al. (2013), who found that snow patch ablation is likely influenced by topography-induced changes in microhabitats. One limitation of the distance to snow/ice patch predictor is that it provides information only on snow presence or absence but not snow depth. However, considering Ayres et al. (2010), who found no significant relationship between soil moisture and snow depth, this limitation may be acceptable for the current approach. The individual impacts of the predictors on the soil moisture, indicated by the Shapley values (Fig.5), suggest that the model learned meaningful relationships. Higher values of the TWI indicate areas that are more likely to accumulate water (Boehner et al., 2001), and the Shapley values reflect this in general. As water accumulation is only possible where moisture sources are present in Taylor Valley, it is plausible that the Shapley Values for the TWI do not show an undisturbed pattern. The indication that soil moisture increases with decreasing distance to snow/ ice patch is consistent with the findings of Ayres et al. (2010) and Gooseff et al. (2003a). Higher potential global radiation and daily insolation time both contribute to predictions of higher soil moisture, which is in line with the findings of Fountain et al. (1999).

290 During predictions within the measuring period, most of the area of interest was located within the model's area of applicability (AOA). Nonetheless, the areas outside the AOA repeatedly appeared as line-shaped polygons (Fig.7). The distance to snow/ice patch predictor was the most important temporally dynamic predictor for the model, and therefore areas outside the AOA

likely correspond to predictor states that were underrepresented or not present in the training data. The polygon pattern of the current area outside the AOA supports this interpretation. The distance to closest snow/ice patch predictor contains the nearest-neighbour information for each snow/ice patch. Therefore, it is partitioning the valley into Voronoi-diagram-like areas (Aurenhammer, 1991). These areas all consist of ‘distance to snow/ice patch’ lines equidistant to the respective snow patch. If a certain range of distances was not present in the ground truth data, these lines are likely to be outside the AOA, producing the observed patterns. Because the influence of the snow-patch on soil moisture is likely to decrease with increasing distance (Ayres et al., 2010; Gooseff et al., 2003a), classifying the snow/ice patch distances into distance categories should be considered for further model refinement and to overcome the present AOA limitation.

The Local Point Density patterns (Fig.8) supported this impression. Furthermore, the LPD indicated, that during 19-20 November 2022, different climatic conditions occurred, as the LPD was unusually low over wide areas of Taylor Valley during this period compared to all other time steps.

The predictions (Fig.7) indicate, that, in line with Barrett et al. (2004), the soil moisture changed spatio-temporally across Taylor Valley, while large areas exhibited dry conditions, as expected. The known high spatial heterogeneity of soil moisture (Bockheim and McLeod, 2008) was also reflected in the predictions. Moreover, the TMS soil moisture measurements showed similar spatial variability, in some cases even at sub-meter scales.

5 Conclusion and outlook

Our results indicated that TMS sensors provide a reliable tool for measuring soil moisture in the extreme environment of the MDVs. Soil moisture differed significantly at both small and landscape scales of Taylor Valley. After assessing the usability of TMS soil moisture measurements as ground truth data in Taylor Valley, we trained a random forest model to spatio-temporally model soil moisture across the valley. The model performance metrics and selected predictors indicated that this modelling technique could be used for representative predictions of soil moisture in this environment. To overcome applicability limitations, further model refinement will be required. Our results can contribute to refining and identifying locations for future sensor networks.

The random forest machine learning model showed promising results and could contribute to understanding ecological processes in Taylor Valley at the landscape scale, especially under typical climatic conditions.

While further investigation of the model’s applicability under extreme climatic conditions would be necessary beforehand, using the model to investigate soil moisture responses to extreme events could contribute to our understanding of ecological dynamics during such events at the landscape scale.

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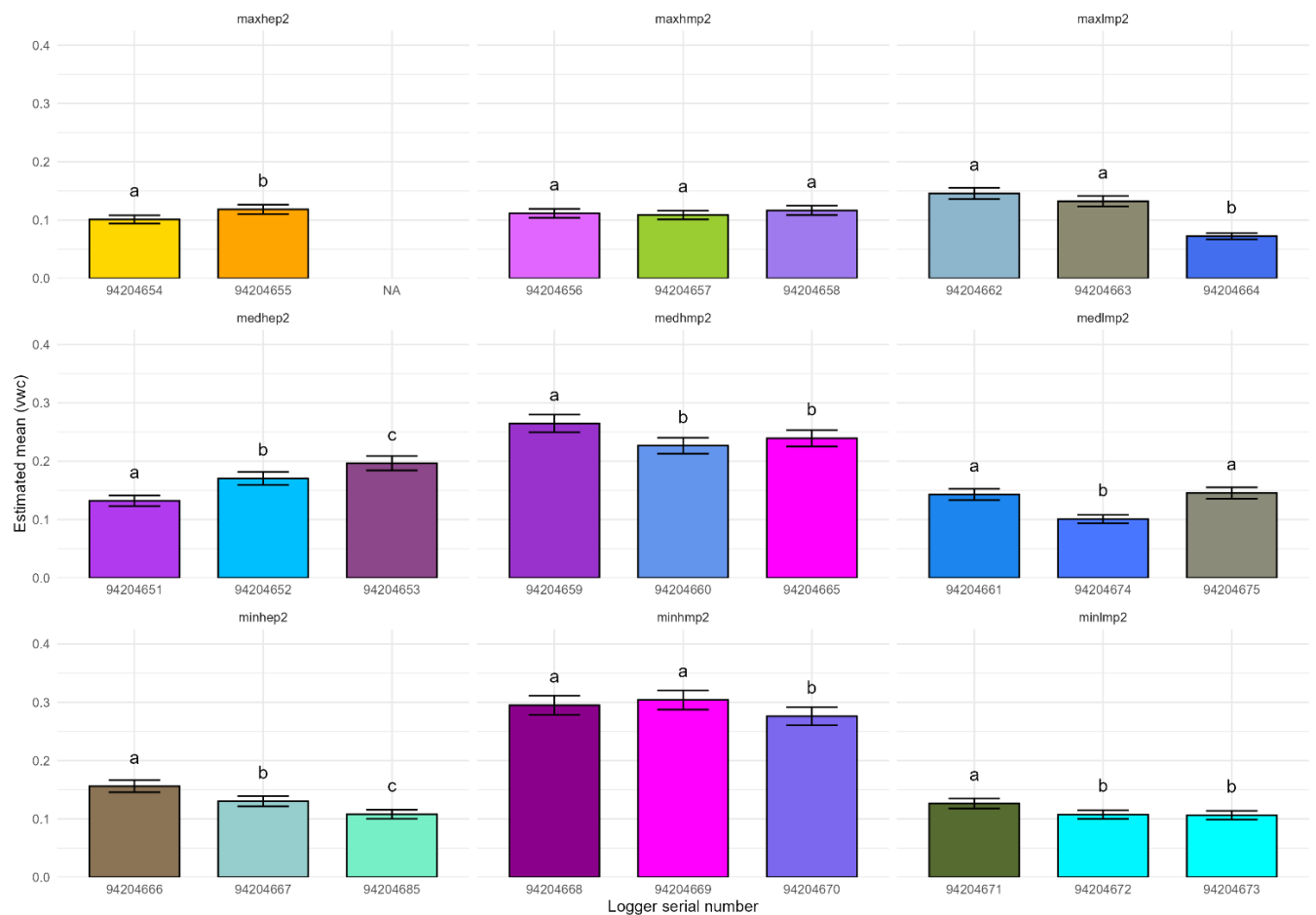
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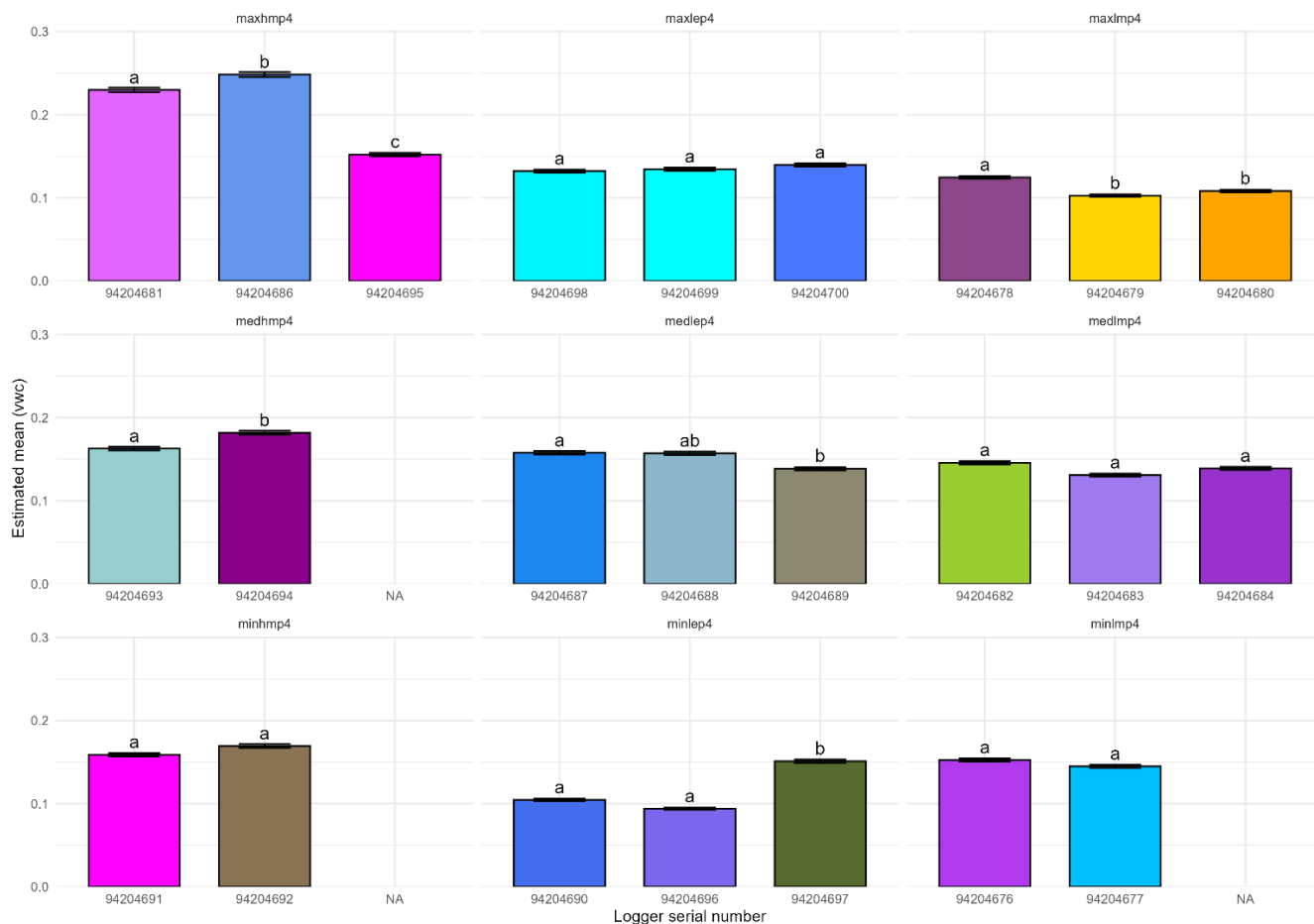
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I Pairwise differences of mean soil moisture between loggers on sub-site level for Plot 2. Identical letters indicate statistical significance ($p < 0.001$) and $|\text{estimate}| > 0.1$ (logger sensitivity). NA indicates that only two loggers were installed on the corresponding site. Colours are logger specific and equal to the colours used for individual loggers in the supplementary material (S1), for better comparability.

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II Pairwise differences of mean soil moisture between loggers on sub-site level for Plot 4. Identical letters indicate statistical significance ($p < 0.001$) and $|\text{estimate}| > 0.1$ (logger sensitivity). NA indicates that only two loggers were installed on the corresponding site. Colours are logger specific and equal to the colours used for individual loggers in the supplementary material (S1), for better comparability.